Graphic Era HILL UNIVERSITY

Established by an Act of the State Legislature of Uttarakhand (Act No. 12 of 2011)
University under section 2(f) of UGC Act, 1956

End Term (Odd) Semester Examination December 2024

Roll no.....

Name of the Course and semester: B.Tech.(ME) 5th Sem
Name of the Paper: Dynamics of Machines
Paper Code: TME 503

Time: 3 hour

Maximum Marks: 100

Note:

(i) All the questions are compulsory.
(ii) Answer any two sub questions from a, b and c in each main question.
(iii) Total marks for each question is 20 (twenty).
(iv) Each sub-question carries 10 marks.

Q1. **(2X10=20 Marks)**

a. Write short notes on the following: CO1
(i) Principle of superposition
(ii) Rigid & Resistant bodies
(iii) Kinematics of machines
(iv) Dynamics of machines

b. (i)What is the principle of Virtual work? Explain. CO1
(ii)What is the principle of superposition as applicable to a system of forces acting on a mechanism. CO1

c. State and explain D'Alembert's principle. CO1

Q2. **(2X10=20 Marks)**

a. Derive the following relations for a horizontal reciprocating engine: CO2
(i) Angular acceleration of the connecting rod
(ii) Acceleration of piston
(iii) Turning moment on crankshaft

b. (i)What do you mean by dynamic equivalent system? Explain. CO2
(ii) What are free-body diagrams of a mechanism? How are they helpful in finding the various forces acting on the various members of the mechanism? CO2

c. Explain turning moment diagram for a **Single-Cylinder** Four -Stroke engine & **Multi -Cylinder** Four -Stroke engine. CO2

Q3. **(2X10=20 Marks)**

a. What is a Clutch? Make a sketch of a Single-Plate Clutch & describe its working. CO3

b. Write short notes on the following: CO3
(i) Coefficient of friction
(ii) Types of friction

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c. Derive the relation for ratio of belt tensions in a flat-belt drive, assuming that the flat-belt is on the point of slipping on the pulleys. CO3

Q4. (2X10=20 Marks)

a A shaft carries four masses A,B,C and D of magnitude 200kg, 300 kg, 400 kg and 200 kg respectively and revolving at radii 80 mm,70 mm, 60 mm and 80 mm in planes measured from A at 300 mm, 400 mm and 700 mm. The angles between the cranks measured anticlockwise are A to B 45^0 , B to C 70^0 and C to D 120^0 . The balancing masses are to be placed in planes X and Y. The distance between the planes A and X is 100 mm, between X and Y is 400 mm and between Y and D is 200 mm. If the balancing masses revolve at a radius of 100 mm, find their magnitudes and angular positions. CO4

b. Derive the following expressions, for an uncoupled two cylinder locomotive engine : CO4
(a) Variation is tractive force (b) Swaying couple (c) Hammer blow.

c. Explain the 'Direct and Reverse Crank' method for determining unbalanced forces in radial engines. CO4

Q5. (2X10=20 Marks)

a. Describe Hartnell type governor with the help of a neat sketch. CO5

b. What do you mean by spin, precession and gyroscopic planes? Derive the expression for the total angular acceleration of the rotor. CO5

c. Define Sensitiveness, Hunting, Isochronism and Stability of a Governor. CO5

Note For the question paper setters:

- Question paper should cover all the COs of the course.
- Please specify COs against each question.